

One of TIL6022's main goals is to prepare you to apply data pipelines to real-world TIL tasks, and to leave you well-qualified to work as a team to tackle quantitative research questions in TIL. The final project is intended to start you in these directions.

There are two main types of projects you can choose from:

1. **Open-ended projects** – you decide the project topic, research questions, and find relevant datasets. The topic should be relevant to TIL (Transport, Infrastructure, and Logistics). To give you some ideas, here are some project examples.
 - a. Application Project
 - E.g., impact of COVID on travel behavior, or evaluating accessibility of public transportation.
 - b. Algorithmic Project
 - E.g., optimization algorithms for logistics, or prediction algorithms for traffic.
 - c. Theoretical Project
 - E.g., car-following model, individual mobility generation model.
2. **Fixed-topic projects** – two project topics related to the 2025 SAIL event in which you apply data analysis and visualization techniques to assist with crowd management. There are two topics:
 - a. Crowd management monitoring dashboard
 - b. Short-term crowd predictions

More detailed project descriptions can be found in Appendix I.

Groups:

The project is done in groups of 5 people; teams are formed by students themselves. Please self-enroll your team in a group in Brightspace (Collaboration -> Groups -> View Available Groups -> Join Group). Detailed instructions on enrolling in a group in Brightspace can be found [here](#). For group collaboration, you will use GitHub to develop the code and work on the Jupyter notebook together. Make one main GitHub repository for your group. Note: you will learn about using GitHub for group collaboration in lecture 8, Versioning Systems.

For fixed topic 2. a, crowd management monitoring dashboard, it will be a joint group project with TIL4030 - TIL Research and Design Methods. So groups selecting this topic need to remain the same for both courses.

Project Deliverables

There are three required deliverables at different phases of your group projects, including the *proposal*, *midterm presentation*, and the *final report*.

The project is graded based on the final report according to the grading rubric (Appendix II). For the proposal and midterm presentation, although they are not graded, they allow us to give you formative feedback on your progress. You will work on one Jupyter notebook for developing your project proposal

and final project report, so consider this Jupyter notebook as a living document. A template of the Jupyter notebook (project_template.ipynb) is provided via Brightspace under Week (5-10) - Projects tab.

Contents of Deliverables

All deliverables should include a title, authors, author contribution statement, introduction, and necessary references. In addition, the following should be included for each deliverable:

- **Proposal** - including objectives, research questions that you are trying to answer, intended data analysis pipeline, list of datasets (and links to data if applicable), and the geographical scale and temporal scale of your study. **Formative feedback** on submissions.
- **Midterm presentation** – prepare a 5-mins presentation and present during your allocated slot in class. The presentation is followed by 5 mins Q&A session for you to receive feedback from lecturers or other students. **Formative feedback** during project hours. The specific slot will be announced later after the group formation.
- **The final report notebook** builds on top of the proposal notebook, including motivation, key visualizations, analysis of results, conclusions, and discussion. **Summative feedback**.

Submission:

The submissions of the project proposal notebook and final reports are via Brightspace. The deadline for each submission is also available in Brightspace. You should submit on Brightspace as a group: that is, for each deliverable, please make one submission for your entire project group and mention your team members. **In the submission, provide a link to the GitHub main repository of your group project.** The repository needs to be made public first before sharing the link. The GitHub repository should include:

- (1) a **README** file to briefly describe your project, how to run your data analysis pipeline, and author information (for final report). *This README file should provide all the necessary information to reproduce your project analysis.*
- (2) your project Jupyter notebook (**.ipynb**) (for both proposal and final report).
- (3) a **PDF or HTML** export of your Jupyter notebook (for both proposal and final report).
- (4) code and visualizations that are important but not included in the Jupyter notebook (for final report).
- (5) a copy of datasets used that can be publicly shared (for final report).
- (6) other items that are important to reproduce your work (for final report).

Note: since this project repository is publicly accessible, please do not include your student ID in any file of the repository.

Evaluation:

- Research question – (10%)
- Coding/Logic - [code blocks] written logic, code description (20%)
- Narrative - [whole project] - (30%)

- Technical quality (30%)
- Collaboration (10%)

Please refer to the Appendix II - TIL Python Project Grading Rubric document for more details.

Reference Materials

Potential data sources

Here are sample data sets

- EUROPE data - <https://ec.europa.eu/eurostat/web/transport/data/main-tables>
- CBS data - https://opendata.cbs.nl/statline/portal.html?_la=en&_catalog=CBS
- Dutch open data - <https://data.overheid.nl>
- Google mobility report - <https://www.google.com/covid19/mobility/>
- COVID data - <https://ourworldindata.org/coronavirus>, <https://data.rivm.nl/covid-19/>
- Flight data - <https://opensky-network.org>
- Public transport data - https://travix.app/?z=12&x=260915.3&y=6251439.5&l=osm_standard&ol=
- Emission data - <https://map.carbonspace.tech>
- Freight data - [2021 Amazon Last Mile Routing Research Challenge Dataset - Registry of Open Data on AWS](#)

Examples of group projects from previous years

You can find examples of the group projects from previous classes in our course GitHub repository:
<https://github.com/TIL6022-Python-Programming/group-projects/blob/main/2025-collection.md>

Appendix I

SAIL Project Topic Descriptions TIL6022

In collaboration with the AiMTT project (<https://aimtt.nl/>)

2026-09-08

Topic 1: Crowd management monitoring dashboard

In the context of increasing urban density and large-scale public events, effective crowd management has become a critical challenge for ensuring safety, optimizing space usage, and improving the overall experience of individuals in shared environments. A real-time monitoring dashboard provides an essential tool for crowd managers and decision-makers, enabling them to visualize crowd movements, and respond proactively to emerging situations. By consolidating data streams into an accessible interface, such a dashboard supports more informed and agile crowd control strategies—improving both operational efficiency and public safety.

SAIL is the largest free nautical event in the world, hosting tall ships from all over the world, moored in the eastern harbour of Amsterdam *IJHaven* (IJ harbour) and across the IJ river. During the 2025 edition taking place from August 20 until August 24, event-specific data, geospatial data, and real-time information on traffic as well as other contexts have been collected as input for crowd management.

Your challenge is to implement a crowd monitoring dashboard for SAIL 2025. This real-time dashboard should contain relevant information for crowd managers, using the data collected during SAIL 2025. This visualisation should be intuitive, helping crowd managers to identify potential (future) problems, improving their decision-making and making it more efficient.

Topic 2: Short-term crowd predictions

With growing urban populations and increasingly crowded public events, managing crowds effectively has become essential to ensure safety, efficient space use, and a positive experience for visitors and local residents. Short-term predictions are an essential element of effective crowd management. By forecasting crowd density and movement patterns, authorities can take timely, preventive actions to avoid congestion and redirect flows where necessary. When integrated into decision-support systems, these predictions enable more informed and dynamic responses, from repositioning staff and signage to adapting infrastructure in the moment. Using AI and data analytics, models can detect behaviour patterns and anticipate surges—such as those triggered by public transport arrivals or breaks during events. Ultimately, short-term predictions act as an early-warning system, allowing operators to maintain control in complex environments and deliver safer, smoother crowd dynamics.

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taking place from August 20 until August 24, event-specific data, geospatial data, and real-time information on traffic as well as other contexts have been collected as input for crowd management.

Your challenge is to provide short term crowd predictions using the data of SAIL 2025. Those predictions may serve as input into a crowd management system or crowd monitoring dashboard.

Materials and Support

Suggested literature:

Daamen, W., Hari, T., Xin, Y., Hoogendoorn-Lanser, S., & Hoogendoorn, S. (2026). Short-term crowd forecasting for SAIL2025. *Acta Polytechnica CTU Proceedings*, 57, 52-60.

Daamen, W., Duives, D., Yuan, Y., & Hoogendoorn, S. (2018). Real-time crowd monitoring dashboard. In *TRB 2018: 97th Annual Meeting of the Transportation Research Board* (pp. 18-03137).

Suggested learning materials:

AiMTT Learning Module — Crowd Management During Events

<https://aimtt.nl/learning-hub/learning-module-crowd-management-during-events/>

Appendix II

(8.5, 10] points	(7.5, 8.5] points	[6, 7.5] points	(0, 6) points	Feedback and comments
CONTENT 100%				
Research Question 10% ☐ The research question is well-formulated, precise, and significant. RQs are well supported by data processing, quantitative analysis, and visualization. RQs are well supported by the data selected.	☐ The research question is clear and relevant. RQ are sufficiently supported by data processing, quantitative analysis, and visualization. RQs are sufficiently supported by the data selected.	☐ The research question is clear. RQ are partially supported by data processing, quantitative analysis, and visualization. RQs are partially supported by the data selected.	☐ The research question is not so clear. RQ are not supported by data processing, quantitative analysis, and visualization. RQs are not supported by the data selected.	
Coding/Logic 20% ☐ Code is exceptionally well-organized and follows best practices. Logic is flawless, and the solution demonstrates a deep understanding of the problem. Code is efficient, maintainable, and thoroughly documented. Demonstrates creative problem-solving and goes beyond the project requirements.	☐ Code is well-structured and follows good coding practices. Logic is sound, and the solution effectively addresses the research question. Code is reasonably efficient, maintainable, and adequately described with comments. Meets most of the project requirements.	☐ Code structure is somewhat disorganized or lacks consistency. Logic contains minor flaws or inefficiencies, but it generally meets the project requirements. Code description is minimal or missing in some areas. Meets the core project requirements without any significant enhancements or creativity.	☐ Code is poorly structured and lacks adherence to coding standards. Logic contains major flaws or errors, making the solution ineffective. Code lacks description, making it difficult to understand. Fails to meet essential project requirements and may not compile or run as expected.	
Technical Quality 30% ☐ Data analysis and visualization is exceptionally well-designed and closely aligns with the project requirements. Utilizes proper modelling techniques and tools. Demonstrates a deep understanding of data relationships and dependencies.	☐ Data analysis and visualization is well-structured and adequately addresses the project requirements. Employs suitable modelling techniques and tools proficiently. Data relationships are logically defined and accurately represent the domain.	☐ Data analysis and visualization is somewhat disorganized or lacks full alignment with the project requirements. Utilizes basic modelling techniques but may have some inaccuracies. Data relationships are represented but may lack some clarity or completeness.	☐ Data analysis and visualization is poorly structured and significantly deviates from project requirements. Utilizes incorrect or inadequate modelling techniques and tools. Data relationships are unclear, incomplete, or incorrect.	
Narrative 30% ☐ The narrative is exceptional, demonstrating a compelling and well-crafted story that engages the reader from start to finish. It effectively sets the stage, clearly defining the project's objectives, significance, and context. The flow of the narrative is seamless, with well-structured paragraphs and logical transitions. The writing style is smooth, demonstrating a mastery of language and conveying complex ideas with ease.	☐ The narrative is proficient, providing a clear and coherent story that effectively communicates the project's purpose and outcomes. It introduces the project with clarity, outlining its objectives and importance. The narrative flows logically, with well-organized sections and appropriate transitions. The writing style is clear and concise, effectively conveying the project's ideas and findings.	☐ The narrative is basic, with some areas needing improvement in clarity and coherence. It introduces the project's objectives but may lack some detail or context. The flow of the narrative is somewhat disjointed, with sections that may not connect seamlessly.	☐ The narrative is unsatisfactory, indicating significant weaknesses in clarity, coherence, and engagement. It may struggle to effectively introduce the project's objectives or establish its significance. The narrative lacks a clear flow, making it difficult for the reader to follow the story. The writing style may be confusing, with issues related to grammar, syntax, or word choice.	

Collaboration 10% ☐	☐	☐	☐	
Consistently uses Git best practices in collaboration (branches, merge, pull requests). Writes detailed commit messages that improve project history. Active participation and excellent coordination within the team; merge conflicts are resolved effectively.	Commits frequently with clear, descriptive messages. Works on feature branches and follows a standard workflow. Sufficient participation and coordination within the team, resolving merge conflicts sufficiently.	Commits code somewhat regularly, but messages are vague or inconsistent. Uses branches but may work directly on main occasionally. Occasionally coordinates with teammates, having minor unresolved merge conflicts.	Rarely commits code or only pushes large, unreviewed changes. Minimal use and understanding of branching/merging. Does not coordinate with teammates, causing merge conflicts without resolving them.	
TOTAL POINTS				
				Final grade
KNOCK OUT CRITERIA				
All group members must participate in the project actively (non-participating members will not receive a mark).				